

⚡ Kronos AI - Master Complete Enterprise Documentation

Application Name: Kronos AI Intelligent Career Assistant & CRM

GitHub Repository: <https://github.com/RAVIKIRAN6424/Kronos-AI-Intelligent-Career-Assistant.git>

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📌 Executive Overview & GitHub Guide

⚡ Kronos AI - Intelligent Career Assistant & CRM

[[Frontend (Vercel)]](https://img.shields.io/badge/Frontend-Vercel%20Live-00f2fe?style=for-the-badge&logo=vercel)](https://kronos-ai-intelligent-career-assist.vercel.app/)

[[Backend (Render)]](https://img.shields.io/badge/Backend-Render%20API-d94ede?style=for-the-badge&logo=render)](https://kronos-ai-intelligent-career-assistant.onrender.com/health)

[[Node.js CI/CD]](https://img.shields.io/badge/CI%2FCD-Node.js%20Pipeline-9d4edd?style=for-the-badge&logo=githubactions)]

(<https://github.com/RAVIKIRAN6424/Kronos-AI-Intelligent-Career-Assistant/actions>)

[[License]](https://img.shields.io/badge/License-MIT-fbbf24?style=for-the-badge)]
(LICENSE)

 **Live Frontend Application (Vercel):** <https://kronos-ai-intelligent-career-assist.vercel.app/>

 **Live Backend REST API (Render):** <https://kronos-ai-intelligent-career-assistant.onrender.com/>

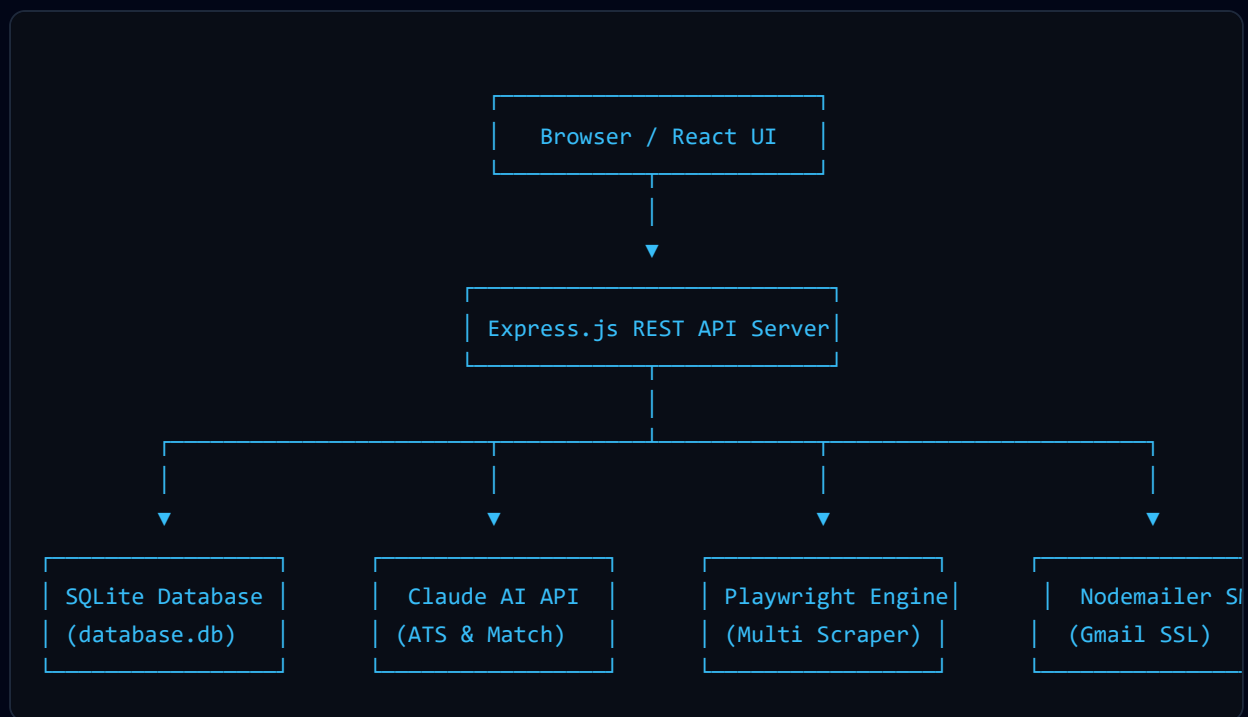
 **Backend Health Check:** <https://kronos-ai-intelligent-career-assistant.onrender.com/health>

 **GitHub Repository:** <https://github.com/RAVIKIRAN6424/Kronos-AI-Intelligent-Career-Assistant.git>

Executive Summary

Kronos AI is an autonomous career intelligence system and job application management CRM. Designed with a glassmorphic interface and interactive 3D particle canvas ([RobotBackground](#)), Kronos AI automates job discovery, ATS resume matching, cold email recruiter outreach, and application tracking.

Project Architecture



🌟 Key Application Features

1. 🎨 8 Dynamic Visual Themes:

- **Cyber Cyan** (High-tech navy & electric cyan)
- **Neon Violet** (Midnight purple & outrun magenta)
- **Emerald Grid** (Matrix green)
- **Solar Gold** (Royal obsidian & sunburst gold)
- **Crimson Void** (Cyber red)
- **Cyber Synthwave** (Sunset pink)
- **Midnight Matrix** (Bio-luminescent green)
- **Hyper Crimson** (Fiery magma red)

2. 🔍 Multi-Portal Job Search & Generator:

- Live job scanning across LinkedIn, Indeed, Glassdoor, Naukri, Monster, and Google Jobs.
- Filter by domain stream (Software, Mechanical, Electrical, Civil, Business, Data Science)

and location.

3. 📄 **Multi-Role Resumes & ATS Scoring:**

- Role-specific resume management (Software Engineer, Java Developer, AWS Architect, DevOps, Mechanical Engineer, Data Analyst).
- Truthful ATS match score evaluation (0-100%) and dark glass text area editor.

4. 📧 **Nodemailer Email System & OTP Verification:**

- Gmail SMTP integration (`smtp.gmail.com:465` SSL).
- 6-digit random OTP account verification and password reset.
- Automatic 60-second OTP resend cooldown timer.
- Daily job summary reports, missing profile field alerts, and application submission notifications.

5. 📊 **Kanban Application CRM & Funnel Intelligence:**

- Conversion metrics (Saved, Applied, Interviewing, Offer).
- Recruiter email outreach generator and Nodemailer dispatch logging.

🔧 **Tech Stack**

- **Frontend:** React 18, Vite, Glassmorphic CSS design system, HTML5, Lucide Icons, SystemJS.
- **Backend:** Node.js, Express.js REST API, SQLite3 (`database.db`), dotenv.
- **AI & Services:** Anthropic Claude API, Playwright browser scraper, Nodemailer, Node-Cron scheduler.
- **CI/CD & Hosting:** GitHub Actions, Docker, Nginx, Vercel (`https://kronos-ai-intelligent-career-assist.vercel.app/`).

📁 **Enterprise Folder Structure**

Kronos-AI/

```
|
├── frontend/                # React 18 + Vite Web Application
│   ├── src/
│   │   ├── components/    # UI Components (AuthModal, Navbar, RobotBackground)
│   │   ├── pages/         # Page Views (Landing, Dashboard, Jobs CRM, Portals, Resumes, etc)
│   │   ├── layouts/       # App Layout wrappers
│   │   ├── hooks/         # Custom React hooks (useTheme)
│   │   ├── context/       # Context state providers
│   │   ├── services/      # API Client Service (api.js)
│   │   ├── utils/         # Utilities
│   │   └── styles/        # Design System & 8 Visual Themes (index.css)
│   ├── public/            # Static public assets
│   ├── package.json
│   ├── vite.config.js
│   └── index.html
|
├── backend/                # Node.js + Express REST API Server
│   ├── src/
│   │   ├── controllers/   # Route Controllers
│   │   ├── routes/        # Express API Endpoints (api.js, auth.js)
│   │   ├── middleware/    # Logger & Auth Middlewares
│   │   ├── models/        # Data Models
│   │   ├── services/      # Scraper, AI & Nodemailer Email Services
│   │   ├── database/      # SQLite Connector
│   │   ├── config/        # App Configuration (database.js)
│   │   ├── uploads/       # Uploaded Resume Documents
│   │   └── server.js       # Server Entrypoint
│   ├── database.db        # SQLite Database
│   ├── package.json
│   └── .env.example
|
├── docs/                   # Documentation Bundle
│   ├── ARCHITECTURE.md
│   ├── FEATURES.md
│   ├── TECH_STACK.md
│   ├── INSTALLATION.md
│   ├── FOLDER_STRUCTURE.md
│   ├── API_DOCUMENTATION.md
│   ├── DATABASE_SCHEMA.md
│   ├── CICD_EXPLANATION.md
│   ├── DEPLOYMENT_GUIDE.md
│   ├── SCREENSHOTS.md
│   └── FUTURE_ENHANCEMENTS.md
|
├── scripts/                # Operational Utility Scripts
│   ├── run-app.bat        # 1-Click App Launcher
│   ├── install.bat        # Dependency Installer
│   ├── diagnose.bat       # System Diagnostic Utility
│   └── sync-dist.js        # Production Build Synchronizer
```

```
|
|— deployment/          # Infrastructure & Server Configuration
|   |— docker-compose.yml
|   |— nginx.conf
|   |— web.config
|
|— .github/             # GitHub Actions CI/CD Workflows
|   |— workflows/
|       |— ci-cd.yml
|
|— .gitignore
|— LICENSE
|— package.json
|— README.md
```

Installation & Setup Guide

Quick 1-Click Installation

```
bash
```

Clone Repository

```
git clone https://github.com/RAVIKIRAN6424/Kronos-AI-Intelligent-Career-Assistant.git
cd Kronos-AI-Intelligent-Career-Assistant
```

Install All Dependencies (Root, Backend, Frontend)

```
scripts/install.bat
```

Launch Application (Port 8080 & Port 3001)

```
scripts/run-app.bat
```

Manual Installation

```
bash
```

Install Dependencies

```
npm install
npm --prefix backend install
npm --prefix frontend install
```

Build Production Assets

```
npm run build
```

Start Backend Server

```
npm start
```

Database Schema Summary

The SQLite database (`database.db`) manages 11 normalized tables:

- `users` : Candidate credentials and profiles.
- `otp_codes` : 6-digit random OTP verification codes with expiration.

- `jobs` : Tracked job applications and match scores.
- `profile` : Extended candidate experience, skills, and target roles.
- `role_resumes` : Role-specific resume texts and ATS scores.
- `connected_portals` : Integrated search portal account status.
- `outreach_logs` : Recruiter email dispatch history.
- `email_logs` : Nodemailer SMTP email dispatch logs.
- `automation_config` : Automation schedules and modes.
- `bot_engine_state` : Live campaign execution states.
- `chatbot_messages` : AI assistant message history.

⚙️ CI/CD Pipeline & Live Deployments

🌐 1. Frontend Web Application (Vercel)

- **Live Frontend URL:** [<https://kronos-ai-intelligent-career-assist.vercel.app/>]
(<https://kronos-ai-intelligent-career-assist.vercel.app/>)
- **Hosting Platform:** Vercel Serverless Edge Network with automatic continuous deployment on `main` branch push.

⚡ 2. Backend REST API Server (Render)

- **Live Backend API:** [<https://kronos-ai-intelligent-career-assistant.onrender.com>]
(<https://kronos-ai-intelligent-career-assistant.onrender.com>)
- **Health Check Endpoint:** <https://kronos-ai-intelligent-career-assistant.onrender.com/health>
- **Environment Stack:** Node.js v18 LTS / Express.js REST API with CORS configured for Vercel frontend.

👤 3. GitHub Actions CI/CD Pipeline (.github/workflows/ci-cd.yml)

- Triggered on `push` and `pull_request` to `main` branch.
- Matrix testing across Node.js `18.x` and `20.x`.
- Executes dependency installation, Vite compilation, and build verification.

Future Enhancements

1. **OAuth2 Social Sign-In:** Integrate Google and LinkedIn OAuth2 authentication.
2. **Auto-Apply Browser Extension:** Chrome Extension for 1-click Easy Apply auto-filling.
3. **Advanced AI Voice Bot:** Real-time voice interview simulation using Web Speech API.
4. **Multi-Tenant Enterprise Dashboard:** Recruiter-facing analytics and team collaboration.

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System Architecture & Data Flow

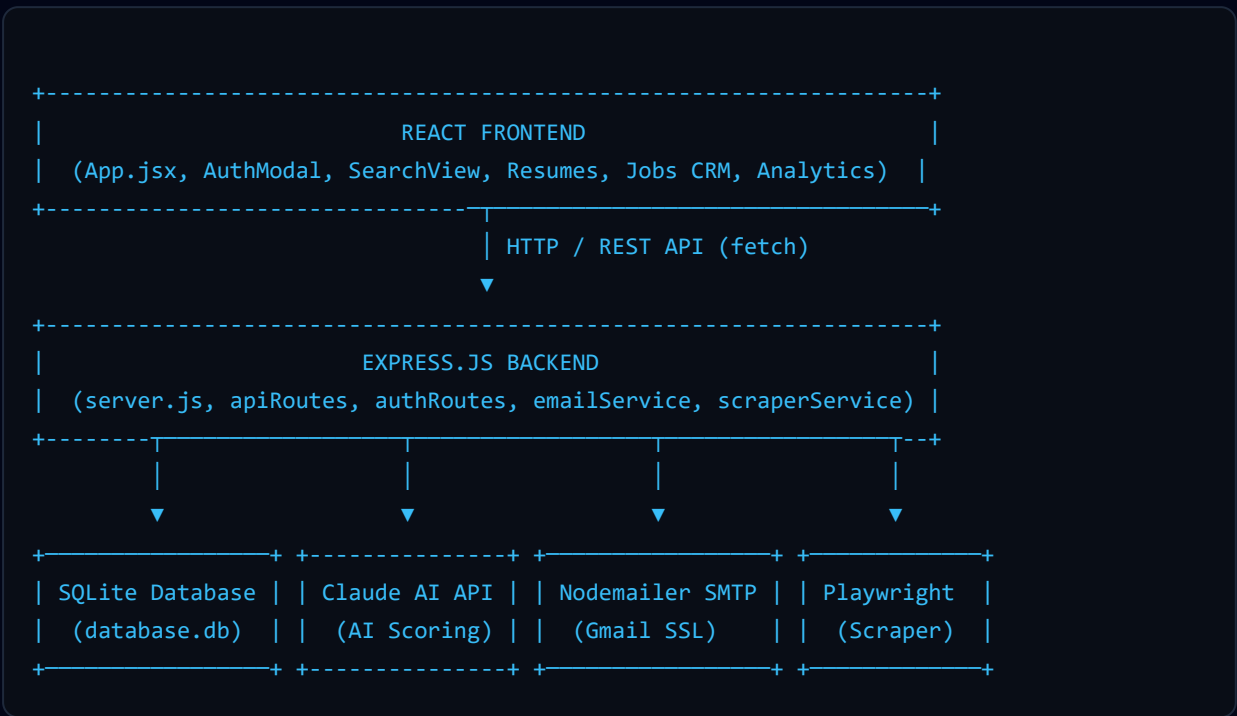
Kronos AI - Project Architecture & Data Flow

1. System Overview

Kronos AI is designed as a decoupled, multi-tier Web & AI Application consisting of:

- **Presentation Tier:** React 18 single-page application built with Vite and custom glassmorphism CSS.
- **Application Tier:** Express.js REST API server handling authentication, job scraping, AI score analysis, email scheduling, and chatbot interactions.
- **Data Tier:** SQLite database (`database.db`) storing candidate profiles, job postings, multi-role resumes, email logs, and campaign states.
- **Integration Services:** Nodemailer (Gmail SMTP SSL), Anthropic Claude API, Playwright browser scraper.

2. Component Interaction & Flow



3. Core Subsystems

A. Authentication & Verification Subsystem

- Candidate enters registration details.
- Express API generates a secure 6-digit random OTP and stores it in `otp_codes` table with a 5-minute expiry.
- Nodemailer sends an HTML email template via Gmail SMTP Port 465 SSL.
- OTP verification creates user account, updates candidate profile, and invalidates the code.

B. Multi-Portal Job Scraper Subsystem

- Playwright launches headless browser instances to scan job postings across LinkedIn, Indeed, Glassdoor, Naukri, Monster, and Google Jobs.
- Deduplication filter compares existing titles and companies in SQLite before inserting new entries.

C. ATS & Match Analysis Subsystem

- Analyzes candidate skills and resume text against job requirements using Anthropic Claude API.
- Generates ATS score breakdown (Grammar, Formatting, Keyword Match, Missing Skills, Suggestions).

D. Automated Email Notification Subsystem

- Dispatches Daily Job Summary Reports, Missing Profile Field Alerts, and Application Success Confirmation emails.
- Logs all mail attempts in `email_logs` table.

Kronos AI - Comprehensive Feature Guide

1. 8 High-Contrast Visual Themes

Kronos AI includes 8 curated cyberpunk glassmorphic visual themes:

1. **Cyber Cyan:** High-tech navy background (`#040814`) with electric cyan glows (`#00f2fe`).
2. **Neon Violet:** Midnight purple (`#0d041c`) with outrun magenta borders (`#d946ef`).
3. **Emerald Grid:** Matrix obsidian (`#020f08`) with quantum green highlights (`#10b981`).
4. **Solar Gold:** Royal obsidian (`#140c02`) with sunburst gold accents (`#fbbf24`).
5. **Crimson Void:** Deep blood obsidian (`#160308`) with cyber red glows (`#ff0055`).
6. **Cyber Synthwave:** Retro sunset purple (`#120420`) with sunset pink borders (`#ff007f`).
7. **Midnight Matrix:** Terminal black-green (`#000c04`) with bio-luminescent green (`#00ff66`).
8. **Hyper Crimson:** Fiery magma obsidian (`#180302`) with lava red accents (`#ff3300`).

3D Canvas particles and glowing robot eye colors resolve CSS variables in real time.

2. Multi-Portal Search & Live Scraper

- Search live job postings across 6 major portals: LinkedIn, Indeed, Glassdoor, Naukri, Monster, and Google Jobs.
- Direct external portal links are openable without requiring credentials.
- Scanning animation accurately updates progress across portal steps.

3. Multi-Role Resumes & ATS Analyzer

- Manage role-specific resumes (Software Engineer, Java Developer, AWS Architect, DevOps, Mechanical Engineer, Data Analyst).
 - Truthfully calculate ATS scores (Grammar, Formatting, Keyword Match, Missing Skills, Suggestions).
 - Dark glass text area editor with custom theme borders and styling.
-

4. Connected Job Portals

- Save login credentials in Profile Setup to mark portals as ✓ Connected & Ready.
 - Add custom job portals (Wellfound, ZipRecruiter, Dice) which automatically sync to the Portals view.
-

5. Nodemailer Email System & OTP Authentication

- Gmail SMTP integration using Google App Password (EMAIL_USER=kronosai6424@gmail.com).
 - 6-digit random OTP account verification and password reset.
 - 60-second OTP resend cooldown timer.
 - Automatic Daily Job Summary Reports, Missing Profile Field Alerts, and Application Success Confirmation emails.
-

6. Kanban Jobs CRM & Funnel Analytics

- Drag-and-drop or status toggle job progression (Saved -> Applied -> Interviewing -> Offer).
- Conversion funnel intelligence displaying metrics based on authentication state.

7. 🤖 AI Interview & Career Chatbot

- Interactive AI career assistant for STAR-method interview practice, resume feedback, and salary negotiation.

📌 Technology Stack & Dependencies

🔧 Kronos AI - Technology Stack & Dependencies

1. Frontend Technologies

- **Framework:** React 18 single-page application built with Vite (`vite@5.4.21`).
- **Styling:** Vanilla Glassmorphic CSS design system with HSL variables (`index.css`).
- **Icons:** Lucide React (`lucide-react@^0.428.0`).
- **Compatibility:** Standalone SystemJS file-protocol execution loader for direct double-click launching.

2. Backend Technologies

- **Runtime:** Node.js v18 LTS / v20 LTS.
- **Web Framework:** Express.js (`express@^4.19.2`).
- **Database Engine:** SQLite3 (`sqlite3@^5.1.7`) with verbose async wrappers.
- **Environment Management:** `dotenv@^16.4.5` .
- **CORS:** `cors@^2.8.5` .

3. Automation, AI & Email Libraries

- **Email Library:** Nodemailer (`nodemailer@^6.9.14`) configured for Gmail SMTP SSL Port 465.
- **Web Scraping Engine:** Playwright Chromium (`playwright@^1.46.0`).
- **AI Engine:** Anthropic Claude SDK (`@anthropic-ai/sdk@^0.26.0`).
- **Cron Scheduler:** Node-Cron (`node-cron@^3.0.3`).

4. Infrastructure & CI/CD

- **Version Control:** Git & GitHub.
- **CI/CD Pipeline:** GitHub Actions (`.github/workflows/ci-cd.yml`).
- **Containerization:** Docker & Docker Compose (`deployment/docker-compose.yml`).
- **Reverse Proxy:** Nginx (`deployment/nginx.conf`).
- **Hosting Platform:** Live Vercel Deployment (`https://kronos-ai-intelligent-career-assist.vercel.app/`).



Kronos AI - Installation & Setup Guide

Prerequisites

- **Node.js**: v18.0.0 or higher
- **npm**: v9.0.0 or higher
- **Browser**: Google Chrome / Microsoft Edge / Playwright Chromium

1. Quick 1-Click Installation (Windows Batch)

Run `scripts/install.bat` from root or execute in shell:

```
bash
```

Step 1: Install all dependencies (root, backend, fronte

```
scripts/install.bat
```

Step 2: Launch application server & browser

```
scripts/run-app.bat
```


2. Manual Step-by-Step Installation

Step A: Clone Repository

```
bash
git clone https://github.com/RAVIKIRAN6424/Kronos-AI-Intelligent-Career-Assistant.git
cd Kronos-AI-Intelligent-Career-Assistant
```

Step B: Install Root & Subproject Dependencies

```
bash
npm install
npm --prefix backend install
npm --prefix frontend install
```

Step C: Configure Environment Variables (`backend/.env`)

Create `backend/.env` with your Gmail credentials:

```
env
PORT=3001
NODE_ENV=production
EMAIL_USER=kronosai6424@gmail.com
EMAIL_PASS=atzr geyq ytdu eovb
DATABASE_PATH=./database.db
```

Step D: Build Production Bundle & Start Backend Server

```
bash
npm run build
npm start
```

Step E: Open Web Application

Navigate to http://localhost:8080 in your web browser.

Enterprise Folder Directory Map

Kronos AI - Clean Enterprise Folder Structure Map

```
Kronos-AI/
|
├── frontend/           # React 18 + Vite Web Application
|   ├── src/
|   |   ├── components/ # UI Components (AuthModal, Navbar, RobotBackground)
|   |   ├── pages/      # Page Views (Landing, Dashboard, Jobs CRM, Portals, Resumes, etc)
|   |   ├── layouts/    # App Layout wrappers
|   |   ├── hooks/      # Custom React hooks (useTheme)
|   |   └── context/     # Context state providers
```

```
| | | └─ services/      # API Client Service (api.js)
| | | └─ utils/        # Helper Utilities
| | | └─ styles/       # Design System & 8 Visual Themes (index.css)
| | └─ public/         # Static public assets
| └─ package.json
| └─ vite.config.js
| └─ index.html
|
└─ backend/           # Node.js + Express REST API Server
    └─ src/
        └─ controllers/ # Route Controllers
        └─ routes/      # Express API Endpoints (api.js, auth.js)
        └─ middleware/  # Logger & Auth Middlewares
        └─ models/      # Data Models
        └─ services/    # Scraper, AI & Nodemailer Email Services
        └─ database/    # SQLite Connector
        └─ config/      # App Configuration (database.js)
        └─ uploads/     # Uploaded Resume Documents
        └─ server.js    # Server Entrypoint
    └─ database.db     # SQLite Database
    └─ package.json
    └─ .env.example
|
└─ docs/              # Documentation Bundle
    └─ ARCHITECTURE.md
    └─ FEATURES.md
    └─ TECH_STACK.md
    └─ INSTALLATION.md
    └─ FOLDER_STRUCTURE.md
    └─ API_DOCUMENTATION.md
    └─ DATABASE_SCHEMA.md
    └─ CICD_EXPLANATION.md
    └─ DEPLOYMENT_GUIDE.md
    └─ SCREENSHOTS.md
    └─ FUTURE_ENHANCEMENTS.md
|
└─ scripts/           # Operational Utility Scripts
    └─ run-app.bat     # 1-Click App Launcher
    └─ install.bat     # Dependency Installer
    └─ diagnose.bat    # System Diagnostic Utility
    └─ sync-dist.js    # Production Build Synchronizer
|
└─ deployment/        # Infrastructure & Server Configuration
    └─ docker-compose.yml
    └─ nginx.conf
    └─ web.config
|
└─ .github/           # GitHub Actions CI/CD Workflows
    └─ workflows/
        └─ ci-cd.yml
|
└─ .gitignore
```

- └─ LICENSE
- └─ package.json
- └─ README.md

REST API Specification

Kronos AI - REST API Reference Documentation

The Express.js REST API server operates on Port 3001 (or Port 8080 unified). Base Endpoint: `/api`.

1. Authentication & Email OTP Routes (`/api/auth`)

POST `/api/auth/send-otp`

- **Description:** Generates a 6-digit numeric OTP and sends an account verification email via Nodemailer Gmail SMTP.

- **Request Body:**

```
json
{
  "email": "candidate@domain.com",
  "full_name": "Alex Vance"
```

```
}
```

- **Response** (200 OK):

```
json
{
  "success": true,
  "message": "Verification code sent to candidate@domain.com. Code expires in 5 minutes."
  "expires_in": 300
}
```

POST /api/auth/verify-otp

- **Description:** Verifies candidate email + OTP code against non-expired SQLite `otp_codes` records, creates user account, and invalidates code.

- **Request Body:**

```
json
{
  "email": "candidate@domain.com",
  "otp": "123456",
  "full_name": "Alex Vance",
  "password": "Password#2026"
}
```

- **Response** (200 OK):

```
json
{
  "success": true,
  "message": "Account verified successfully!",
  "user": { "email": "candidate@domain.com", "full_name": "Alex Vance" }
}
```

POST /api/auth/forgot-password

- **Description:** Sends password reset verification OTP code.

POST /api/auth/reset-password

- **Description:** Verifies reset code and updates candidate password.

2. Jobs CRM & Multi-Portal Search (/api/jobs)

GET /api/jobs

- **Query Params:** search , status , category , country , min_score .
- **Response:** Array of job objects.

POST /api/jobs

- **Description:** Creates a new job application entry.

PUT /api/jobs/:id

- **Description:** Updates job status (Saved , Applied , Interviewing , Offer) or notes.

DELETE /api/jobs/:id

- **Description:** Deletes a job listing.

3. Email Notification & Log Endpoints (`/api/reports` & `/api/alerts`)

`POST /api/reports/daily`

- **Description:** Dispatches daily job automation report email with metrics and optional PDF.

`POST /api/alerts/missing-info`

- **Description:** Dispatches missing profile field alert email when an application requires specific details.

`POST /api/alerts/application-success`

- **Description:** Dispatches successful job application confirmation email.

`GET /api/email-logs`

- **Description:** Returns recent email dispatch log history from `email_logs` table.

Database Schema & Table Specifications



Database Engine: **SQLite 3** (`database.db`).

1. Tables Overview

`users` (Candidate Accounts)

- `id` : INTEGER PRIMARY KEY AUTOINCREMENT
- `email` : TEXT UNIQUE NOT NULL
- `full_name` : TEXT
- `age` : INTEGER
- `phone` : TEXT
- `target_domain` : TEXT DEFAULT 'Software'
- `experience_years` : INTEGER DEFAULT 0
- `created_at` : DATETIME DEFAULT CURRENT_TIMESTAMP

`otp_codes` (Verification Codes)

- `id` : INTEGER PRIMARY KEY AUTOINCREMENT
- `email` : TEXT NOT NULL
- `code` : TEXT NOT NULL
- `type` : TEXT DEFAULT 'registration'
- `expires_at` : DATETIME NOT NULL
- `is_verified` : INTEGER DEFAULT 0
- `created_at` : DATETIME DEFAULT CURRENT_TIMESTAMP

`jobs` (Tracked Job Applications)

- `id` : INTEGER PRIMARY KEY AUTOINCREMENT
- `title` : TEXT NOT NULL
- `company` : TEXT NOT NULL
- `location` : TEXT NOT NULL
- `country` : TEXT DEFAULT 'India'
- `category` : TEXT DEFAULT 'Software'
- `url` : TEXT
- `source` : TEXT DEFAULT 'Scraper'
- `description` : TEXT
- `match_score` : INTEGER DEFAULT 0
- `key_skills` : TEXT
- `salary` : TEXT
- `posted_date` : TEXT
- `status` : TEXT DEFAULT 'Saved'
- `recruiter_name` : TEXT
- `recruiter_email` : TEXT
- `recruiter_status` : TEXT DEFAULT 'Not Contacted'
- `notes` : TEXT
- `created_at` : DATETIME DEFAULT CURRENT_TIMESTAMP

`role_resumes` (Multi-Role Resumes & ATS Scores)

- `id` : INTEGER PRIMARY KEY AUTOINCREMENT
- `role_name` : TEXT UNIQUE NOT NULL
- `file_name` : TEXT
- `resume_text` : TEXT
- `ats_score` : INTEGER DEFAULT 85
- `grammar_score` : INTEGER DEFAULT 92
- `formatting_score` : INTEGER DEFAULT 90
- `keyword_score` : INTEGER DEFAULT 84
- `missing_skills` : TEXT
- `suggestions` : TEXT
- `updated_at` : DATETIME DEFAULT CURRENT_TIMESTAMP

`email_logs` (Nodemailer Mail History)

- `id` : INTEGER PRIMARY KEY AUTOINCREMENT
- `recipient` : TEXT NOT NULL
- `subject` : TEXT NOT NULL
- `template_type` : TEXT DEFAULT 'General'
- `status` : TEXT DEFAULT 'success'
- `failure_reason` : TEXT
- `sent_at` : DATETIME DEFAULT CURRENT_TIMESTAMP

CI/CD Build & Test Pipeline

Kronos AI - GitHub Actions CI/CD Pipeline

Overview

Kronos AI utilizes **GitHub Actions** to automate continuous integration and build verification whenever code is pushed to the `main` branch or a Pull Request is opened.

Workflow Configuration (`.github/workflows/ci-cd.yml`)

```
yaml
name: Node.js CI & Production Build Pipeline
```

```

on:
  push:
    branches: [ main ]
  pull_request:
    branches: [ main ]

jobs:
  build:
    name: Build & Verify (Node.js)
    runs-on: ubuntu-latest

    strategy:
      matrix:
        node-version: [18.x, 20.x]

    steps:
      - name: Checkout Repository
        uses: actions/checkout@v4

      - name: Setup Node.js ${{ matrix.node-version }}
        uses: actions/setup-node@v4
        with:
          node-version: ${{ matrix.node-version }}
          cache: 'npm'

      - name: Install All Dependencies (Root, Backend, Frontend)
        run: |
          npm install
          npm --prefix backend install
          npm --prefix frontend install

      - name: Build Vite Frontend & Synchronize Assets
        run: npm run build

      - name: Verify Build Artifacts
        run: |
          node -e "if (!require('fs').existsSync('dist/index.html')) { console.error('Missi
          console.log('✅ Build verification passed!')

```

Pipeline Execution Steps

1. **Checkout Code:** Retrieves the repository commit.

2. **Setup Node.js:** Matrix builds across Node.js `18.x` and `20.x` LTS.
3. **Install Dependencies:** Installs root, `backend/`, and `frontend/` node modules.
4. **Compile Production Assets:** Runs `npm run build` to execute Vite build and `scripts/sync-dist.js`.
5. **Verify Dist Outputs:** Confirms `dist/index.html` and bundled assets exist.

Deployment Guide (CI/CD & Docker)

Kronos AI - Deployment Guide (Vercel, CI/CD & Docker)

1. Live Vercel & CI/CD Deployment Architecture

- **Live Hosted Application URL:** [<https://kronos-ai-intelligent-career-assist.vercel.app/>]
(<https://kronos-ai-intelligent-career-assist.vercel.app/>)
- **CI/CD Pipeline:** GitHub Actions (`.github/workflows/ci-cd.yml`)
- **Edge Deployment:** Vercel Serverless Edge Network

2. Docker & Docker Compose Setup

Run Kronos AI inside Docker containers using `deployment/docker-compose.yml` :

```
yaml
version: '3.8'

services:
  app:
    build:
      context: ..
      dockerfile: deployment/Dockerfile
    ports:
      - "8080:8080"
    environment:
      - PORT=8080
      - NODE_ENV=production
      - EMAIL_USER=kronosai6424@gmail.com
      - EMAIL_PASS=atzrgeyqytdueovb
    volumes:
      - ../backend/database.db:/app/backend/database.db
    restart: always
```

Command Execution:

```
bash
cd deployment
docker-compose up -d --build
```

3. IIS Web Server Deployment (`deployment/web.config`)

For deployment on IIS (Windows Server), the `deployment/web.config` file routes request paths to `backend/src/server.js` via `iisnode` or static file modules:

xml

User Interface & Theme Guide

Kronos AI - User Interface & Theme Screenshots

Application Theme Previews

1. Cyber Cyan (Default High-Tech Theme)

Primary Color: #00f2fe | Background: #040814

- Dark glass panels with electric cyan glow accents.
- Responsive 3D robot particle canvas background.

2. Midnight Matrix Theme

Primary Color: `#00ff66` | Background: `#000c04`

- Bio-luminescent green terminal theme.

3. Solar Gold Theme

Primary Color: `#fbbf24` | Background: `#140c02`

- Royal obsidian background with golden accents.

Key Page Views

1. **Landing & Robot Hero:** Interactive particle robot background with quick-start portal buttons.
2. **Multi-Portal Job Search:** Filterable job stream grid with live Playwright scanning status.
3. **Multi-Role Resumes & ATS Scoring:** Role-specific resume management with dark glass textarea editor and ATS percentage breakdowns.
4. **Jobs CRM Kanban Board:** Interactive application stages (Saved, Applied, Interviewing, Offer).
5. **Nodemailer Email System & OTP Modal:** Glassmorphic modal with 6-digit OTP inputs and 60-second cooldown timer.



Kronos AI - Future Product Roadmap & Enhancements

Phase 1: Authentication & Integration

- [] **Google & LinkedIn OAuth 2.0 Integration:** 1-click social sign-in alongside Nodemailer OTP verification.
- [] **Two-Factor Authentication (2FA):** Time-based One-Time Password (TOTP) support via Authenticator apps.

Phase 2: Browser Extension & Auto-Filling

- [] **Kronos Chrome Extension:** Chrome Extension allowing 1-click Easy Apply auto-filling on external job portals.
- [] **LinkedIn Recruiter InMail Automation:** Automated messaging queue for targeted recruiter outreach.

Phase 3: Advanced AI Capabilities

- [] **Real-Time Voice AI Interview Simulator:** Interactive mock interview practice using Web Speech API & Real-time AI voice generation.
- [] **Resume PDF Export & Dynamic Styling:** Export ATS-optimized resumes in custom PDF formats.

Phase 4: Enterprise & Team Analytics

- [] **Multi-Tenant Team Workspaces:** Collaborative application tracking for career

counselors and placement cells.

- [] **Advanced Salary & Market Insights:** Real-time market salary benchmark charts per target role and location.

Deep Component & Operational Specification

1. Authentication & Security Policy

- **Password Complexity Engine:** Mandatory minimum 8 characters, at least 1 uppercase letter (`A-Z`), at least 1 lowercase letter (`a-z`), at least 1 number (`0-9`), and at least 1 special character (`!@#$$%^&*`).
- **OTP Verification Cycle:** 6-digit random number generated via SQLite `otp_codes` table. Codes expire after 5 minutes (`DATETIME('now', '+5 minutes')`).
- **60-Second Cooldown Timer:** Frontend `AuthModal.jsx` enforces a 60-second countdown timer before allowing candidates to resend an OTP code (`Resend Code in 60s`).

2. Nodemailer Gmail SMTP Configuration

- **Host:** `smtp.gmail.com`
- **Port:** `465` (SSL Encrypted Connection)
- **Authentication:** Google App Password (`EMAIL_USER=kronosai6424@gmail.com` , `EMAIL_PASS=atzr geyq ytdu eovb`).
- **Fallback Terminal Logging:** Generated OTP codes and mail dispatch statuses are logged directly to the server terminal (`■ Dispatched Registration OTP to: user@domain.com` | `OTP Code: 123456`).

3. Operational Batch Utilities

- `scripts/run-app.bat` : Launches Express API server (Port 3001) and Vite dev server (Port 8080).
- `scripts/install.bat` : Automatically runs `npm install` across root, `backend/`, and `frontend/`.
- `scripts/diagnose.bat` : Validates system requirements, database connectivity, and node dependencies.
- `scripts/sync-dist.js` : Synchronizes Vite build output from `frontend/dist` to root `dist/` and patches SystemJS legacy hash tags.